

Estimating dispersion

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EDMs and GLMs

We say that a distribution belongs to the **exponential dispersion model** family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

or:

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

(dispersion model form)

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$
$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Estimating β

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$
$$g(\mu_i) = \mathbf{x}_i^T \beta$$

Newton's method:

$$\blacktriangleright U(\beta) = \frac{1}{\phi} \sum_{i=1}^n (Y_i - \mu_i) \mathbf{x}_i = \frac{1}{\phi} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu})$$

$$\blacktriangleright \mathbf{H}(\beta) = -\frac{1}{\phi} \sum_{i=1}^n V(\mu_i) \mathbf{x}_i \mathbf{x}_i^T$$

$$\blacktriangleright \mathcal{I}_1(\beta) = \frac{1}{\phi} \mathbb{E}[V(\mu_i) \mathbf{x}_i \mathbf{x}_i^T]$$

we don't need ϕ to estimate β : $\frac{1}{\phi} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}) = 0$

EX: linear reg. $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \Rightarrow \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu}) = 0$

we do need ϕ for inference: $\text{Var}(\hat{\beta}) \approx \frac{1}{n} \mathcal{I}_1^{-1}(\beta)$

EX: linear reg. $\text{Var}(\hat{\beta}) \approx \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$

Estimating ϕ

Dispersion model form:

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

• For some distributions, ϕ is known (e.g. Binomial, Poisson $\phi=1$)

- But still might want to estimate ϕ to check if model is a good fit

• For most distributions, $b(y, \phi)$ is messy — makes it harder to estimate ϕ !

Approximating $b(y, \phi)$

motivating example: Poisson in dispersion model form

$$\begin{aligned} f(y; \mu, \phi) &= \frac{\mu^y e^{-\mu}}{y!} \\ &= \underbrace{\frac{\exp\{y \log(\mu) - y\}}{y!}}_{b(y, \phi)} \exp\left\{-\frac{d(y, \mu)}{2}\right\} \end{aligned}$$

Stirling's formula:

For large y

$$y! \approx y^y e^{-y} \sqrt{2\pi y} = \exp\{y \log y - y\} \sqrt{2\pi y}$$

$$\begin{aligned} \Rightarrow f(y; \mu, \phi) &\approx \frac{1}{\sqrt{2\pi y}} \exp\left\{-\frac{d(y, \mu)}{2}\right\} \\ &= \frac{1}{\sqrt{2\pi \phi v(y)}} \exp\left\{-\frac{d(y, \mu)}{2\phi}\right\} \end{aligned}$$

Saddlepoint approximation

For any EDM,

$$f(y; \mu, \phi) \approx \frac{1}{\sqrt{2\pi\phi v(y)}} \exp\left\{-\frac{a(y, \mu)}{2\phi}\right\}$$

For Normal: $\phi = \sigma^2$ and $v(y) = 1$

(saddlepoint approx. is exact)

For other EDMs, need some conditions for saddlepoint approx. to be good.

Dunn & Smyth suggest:

- Poisson: $y \geq 3$
- Binomial(μ, n): $ny \geq 3$ and $n(1-y) \geq 3$
- Gamma: $\phi \leq \frac{1}{3}$

MLE for ϕ and the mean deviance estimator

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\} \approx \frac{1}{\sqrt{2\pi\phi V(y)}} \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

$$(x_1, y_1), \dots, (x_n, y_n) \quad \text{i.i.d.} \quad y_i | x_i \sim \text{EDM}(\mu_i, \phi)$$

$$\begin{aligned} \mathcal{L}(\phi | x, y, \beta) &\stackrel{\text{approximately}}{\propto} \prod_{i=1}^n \frac{1}{\sqrt{2\pi\phi V(y_i)}} \exp \left\{ -\frac{d(y_i, \mu_i)}{2\phi} \right\} \\ &= (2\pi\phi)^{-\frac{n}{2}} \left(\prod_{i=1}^n V(y_i)^{-1} \right) \exp \left\{ -\frac{1}{2\phi} \sum_{i=1}^n d(y_i, \mu_i) \right\} \end{aligned}$$

$$\mathcal{L}(\phi | x, y, \beta) = -\frac{n}{2} \log(\phi) - \frac{1}{2\phi} \sum_{i=1}^n d(y_i, \mu_i) + \text{constant}$$

$$\frac{d\mathcal{L}}{d\phi} = -\frac{n}{2\phi} + \frac{1}{2\phi^2} \sum_{i=1}^n d(y_i, \mu_i) \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \hat{\phi}_{MLE} = \frac{1}{n} \sum_{i=1}^n d(y_i, \hat{\mu}_i)$$

MLE for ϕ and the mean deviance estimator

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\} \approx \frac{1}{\sqrt{2\pi\phi V(y)}} \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

$$\hat{\phi}_{MLE} = \frac{1}{n} \sum_{i=1}^n d(y_i, \hat{\mu}_i)$$

Biased: $E[\hat{\phi}_{MLE}] = \frac{n-p}{n} \phi \quad (\beta \in \mathbb{R}^p)$

mean deviance estimator: $\hat{\phi}_D = \frac{1}{n-p} \sum_{i=1}^n d(y_i, \hat{\mu}_i)$

Est: $N(\mu_i, \sigma^2) \quad \hat{\sigma}_D^2 = \frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{\mu}_i)^2$

Pearson estimator

$$Y_i | X_i \sim \text{EDM}(\mu_i, \theta)$$

$$\text{var}(Y_i | X_i) = \theta v(\mu_i)$$

$$\Rightarrow \mathbb{E}[(Y_i - \mu_i)^2 | X_i] = \theta v(\mu_i)$$

$$\Rightarrow \mathbb{E}\left[\frac{(Y_i - \mu_i)^2}{v(\mu_i)} | X_i\right] = \theta$$

$$\mathbb{E}\left[\frac{(Y_i - \mu_i)^2}{v(\mu_i)}\right] = \theta$$

Pearson estimator :

$$\hat{\theta}_P = \frac{1}{n-p} \sum_{i=1}^n \frac{(Y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}$$

Ex : $N(\mu_i, \sigma^2)$: $\hat{\sigma}_P^2 = \frac{1}{n-p} \sum_{i=1}^n (Y_i - \hat{\mu}_i)^2$
 $v(\mu_i) = 1$

Activity

Work on the activity (course website).